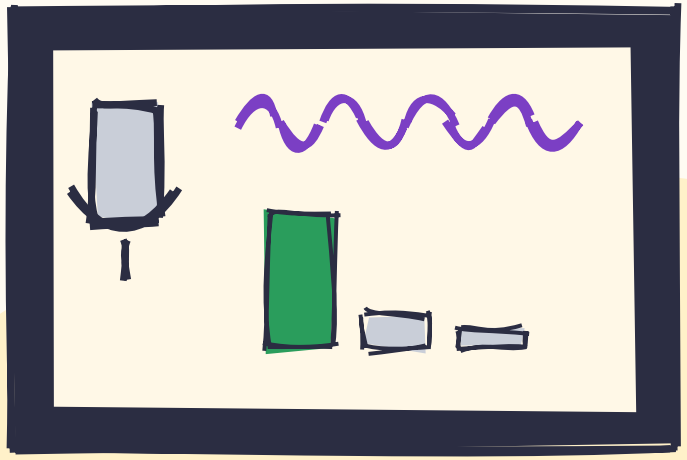
Little Explorers · Class 29

Teach the Computer Our Sounds

Clap, whistle, snap… can the computer tell them apart?

Examples Learning Testing



Did you know?
Phones learned to understand voices by listening to millions of examples!

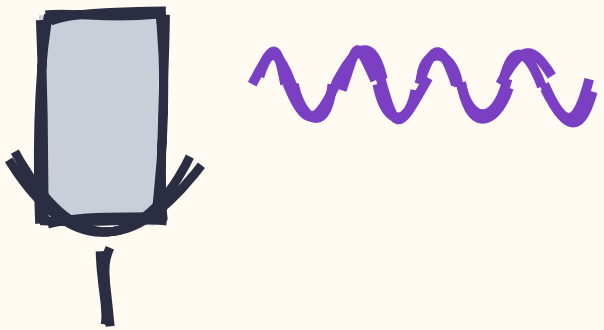
LEARN

Let's wonder...

How does a speaker hear its name?

Hey, computer!

You say 'Hey!' and the smart speaker wakes up. Nobody told it every way people say 'Hey'. So how does it **know**?



It's magic?

Someone listens?

It learned!

Today: you teach a computer to tell a **clap** from a **whistle** from **quiet**, using only examples.

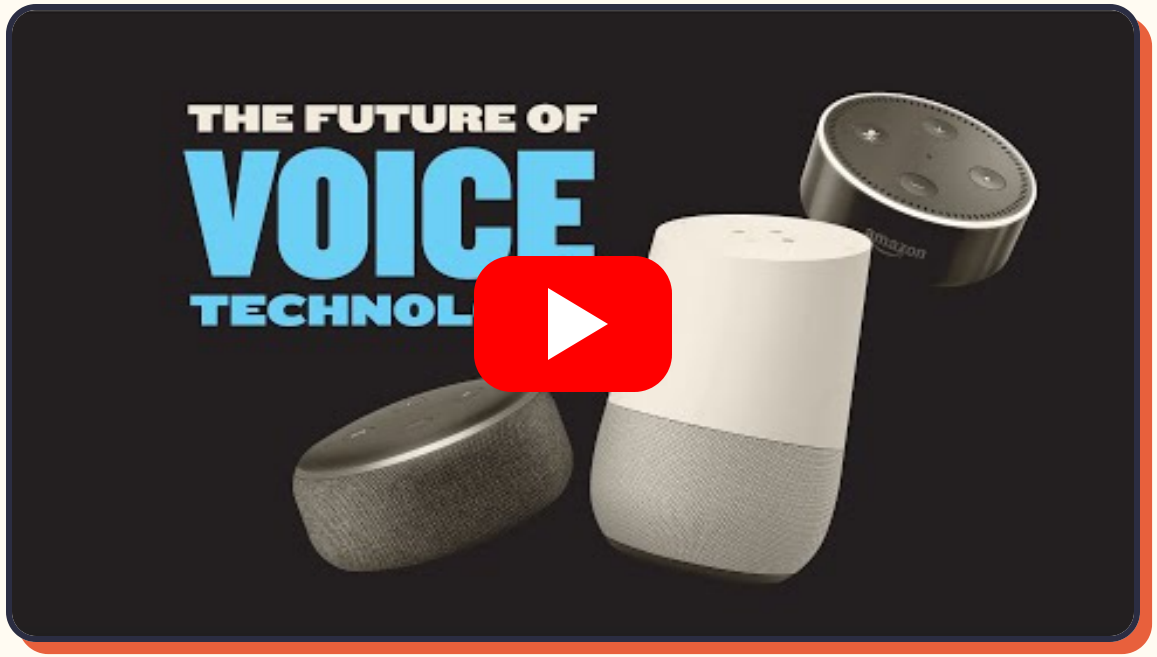
Learning objectives

Today I can...

1	**Sound examples:** give the computer **lots of examples** of each sound	**Proof:** I recorded clap examples
2	**Computer learns:** say that the computer **learns from examples**, not from rules	**Proof:** I explained how it learned
3	**Test it:** **test** the computer and find a sound it gets wrong	**Proof:** I found a mistake and fixed it

Objective 2 · Computer learns · video

Video time!



How Voice Recognition Works

iluli by Mike Lamb · about 5 minutes · youtu.be/2RRT1YuyBCo

Watch for...

1. How does a computer **hear** your voice?
2. Why does it need lots of **examples**?
3. Where do we talk to computers?

Scan to watch

After the video: Clap, whistle and snap. Could a computer learn the difference?

Objective 1 · Sound examples

Every sound has a shape

The idea

When a sound reaches the microphone, the computer sees it as a **wiggly line**.

A clap is one big **spike**. A whistle is a long smooth **wave**. Quiet is almost **flat**!

Example: drawing sounds



Close your eyes and draw in the air while the teacher claps, then whistles. Your hand draws a spike, then a long wave!

Objective 2 · Computer learns

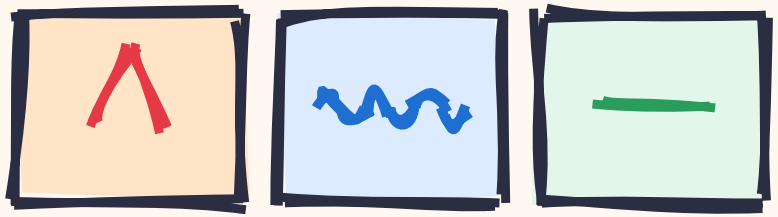
It learns from examples

The idea

We don't tell the computer rules. We give it **lots of examples**: 'this is a clap', 'this is a clap', 'this is a whistle'...

It finds what is the **same** in each group. More examples = smarter computer!

Example: learning animal sounds



clap x10 | whistle x10 | quiet x10

A baby hears 'moo' and sees a cow again and again. Soon the baby knows: moo = cow! Nobody gave the baby a rule.

Real world

Computers that learn sounds



Smart speakers
Wake up when you say their name



Phone typing
Turn your voice into words



Bird apps
Tell which bird is singing



Doctors' helpers
Listen to coughs and heartbeats

? Which sound would YOU teach a computer?

Lots of examples
Voice helpers listened to millions of people so they understand many accents!

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human sound sorter
2. Collect sound examples
3. Train the computer
4. Test and fix
5. Trick the computer
6. Sound scientist challenges

MAKE

Class game

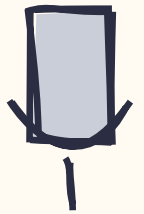
Human sound sorter

How to play

1. One child is the **computer**: eyes closed.
2. The teacher plays 3 sounds, saying the names: **clap, whistle, drum**.
3. Now the teacher plays one sound, with no name.
4. The 'computer' guesses! Swap after 3 turns.

Then discuss

How did the 'computer' know which sound it was?



Big idea

The computer guesses using the examples it heard before.

MAKE

Objective 1 · Build step 1 of 3

Collect sound examples

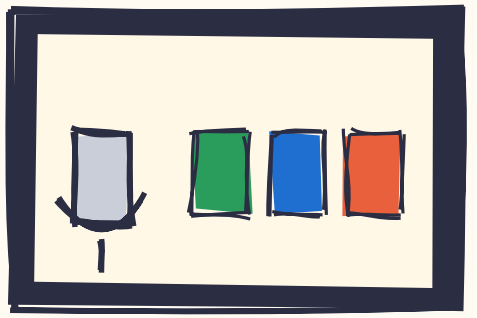
a Teacher opens **Teachable Machine** → Audio project.

b First record **Background noise**: everyone stays quiet for 20 seconds.

c Class 2: name it **Clap**. Children take turns clapping: record 8 or more.

d Class 3: name it **Whistle** (or 'Snap'). Record 8 or more.

e Lots of different children = better examples!



Check: You have 3 groups full of examples.

Keep the room quiet while someone records.

Objective 2 · Build step 2 of 3

Train the computer

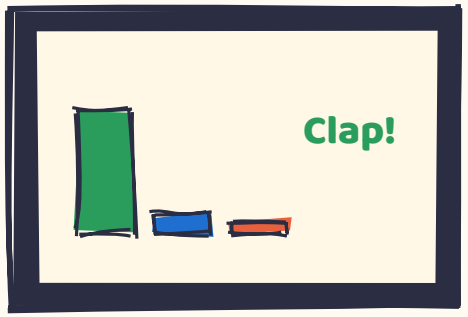
a Click **Train Model**. Wait: the computer is studying!

b When it's done, the **preview** starts listening.

c Clap! Watch the Clap bar go up.

d Whistle! Watch the Whistle bar.

e Stay quiet. Which bar wins?



Check: The right bar goes up for each sound.

Objective 3 · Build step 3 of 3

Test and fix

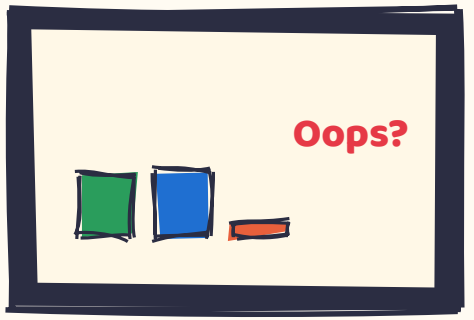
a Try to **trick** it: a soft clap, a very fast clap, a hum.

b Did it make a mistake? That's OK!

c Fix it: record **more examples** of the tricky sound.

d Train again and test again.

e Is it better now?



Check: You found a mistake, added examples, and it got better.

MAKE

Test it!

Trick the computer

Sound I tried	Computer said	Right or wrong?
Soft clap		
Hum		
Snap		

Talk about it

Which sound tricked the computer? How did we fix it?

More examples

When the computer is wrong, it needs more examples, not a telling-off!

Level up!

Sound scientist challenges

Choose your challenge

- **New sound:** add a 4th class, like a drum or a bell.
- **Voices:** teach it 'yes' and 'no' said by many children.
- **Fewer examples:** what happens with only 2 examples each?
- **Noisy room:** does it still work when the class is loud?

Think like an engineer

Why might the computer get confused by a new child's voice?

Boss level

Teach it your favourite animal sound!

Objective check

Can I really do it? Prove it.

01 How did we teach the computer? ○ ○ ○

02 Why do we need lots of examples? ○ ○ ○

03 Show a sound that tricked it, and how you fixed it. ○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Sound show: each team plays 3 sounds and the class cheers when the computer gets it right!

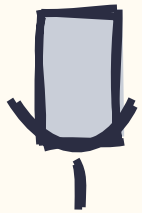
Show what you know

What did we learn?

- [x] Sounds have different shapes
- [x] Computers learn from examples
- [x] More examples = fewer mistakes
- [x] Test it to find mistakes, then fix them

At home

Ask a grown-up which gadgets at home listen to voices.



Next class: Follow-Me Robot Pet!

Exit ticket: why do we record lots of claps, not just one?